

Mealvana Macro Algorithm V3

Edge Function: `generate-macros-v3/index.ts` | March 2026 | Rachel/Xuan corrected

1. BEFORE Workout

Carbs = `weight_kg x min(hours_before, 4.0)` g/kg, minimum 0.5 g/kg [R1, Kerksick 2017]. Minimum 0.5 floor: **NO JUSTIFICATION**

Intensity does NOT change the g/kg – it only suggests an ideal timing window. [R2]

Hours Before	Type	Carbs	Protein	Fat	Hydration
>= 2.5h	Full Meal	wt x hrs g/kg [R1]	wt x 0.25 g/kg NO JUST.	wt x 0.4 g/kg [Blog: 0.3-0.5]	wt x 6.5 ml/kg [Sawka 07: 5-7]
1.0-2.5h	Snack	wt x hrs g/kg [R1]	wt x 0.15 g/kg NO JUST.	5g [Blog: <10g]	wt x 5.5 ml/kg NO JUST.
< 1.0h	Top-up	wt x hrs g/kg [R1]	0g	0g	250 ml NO JUST.
Fasted	None	0g	0g	0g	0

Pre-workout sodium – NO JUSTIFICATION for this entire section. No research source supports these specific values or formulas. [Baker 2017] provides sweat sodium concentration ranges (200-2000 mg/L) during exercise, but no guidance on pre-workout sodium intake amounts.

Base by sweat category: Low=300, Med=450, High=600 mg. Hot environment adds +100mg.

Window	Sodium	Included When
Meal	base + env_bump	Full meal only
Snack	(base + env_bump) x 0.5	Snack + Full meal
Top-up	env_bump + 100	All types

Total = sum of included windows. Accumulates because athlete eats multiple times.

2. DURING Workout

Carbohydrates

Step 1 – Duration band (absolute g/hr, NOT body-weight dependent) [R3, Jeukendrup 2014]:

Duration	Band (g/hr)	Duration	Band (g/hr)
< 60 min	0 - 30	150 - 240 min	60 - 90
60 - 90 min	30 - 60	> 240 min	80 - 100
90 - 150 min	45 - 60		

These bands appear in our [Blog Table] and are informed by [Jeukendrup 2014] and [Kerksick 2017], but **the exact breakpoint values (why 45-60 and not 40-65, etc.) are Mealvana's own creation.** The underlying research gives general guidelines (e.g., “30-60 g/hr”, “up to 90 g/hr”) which we synthesized into these specific 5-band ranges.

Step 2 – Multiply entire band by gut training factor [R4, Jeukendrup 2017, Costa 2019]: Low = 0.7x | Moderate = 1.0x | High = 1.2x

Step 3 – Target = midpoint of scaled band. **NO JUSTIFICATION** – [Rachel] recommended displaying as a range; we chose midpoint for our food selection algorithm.

Step 4 – Cap at sport ceiling [Blog, Pfeiffer 2012]:

Sport	Ceiling	Why Different from Running	Source
Running	70 g/hr	Baseline. Vertical impact causes mechanical GI jostling, limiting tolerance.	[Blog: 50-70, Pfeiffer 2012]
Cycling	120 g/hr	Stable seated position eliminates impact-related GI stress; athletes can tolerate ~70% more carbohydrate.	[Blog: 80-120, Pfeiffer 2012]
Swimming	0 g/hr	Cannot eat or drink during the activity. All fueling shifts to before-phase or brick transitions (§4).	[Blog]

Steps 1-3 are identical for all three sports — same duration bands, same gut training multipliers, same midpoint selection. The sport ceiling is the **only** differentiator. This means gut-trained cyclists on long rides can realize carb rates that would be capped for runners.

Example (run): 2hr run, moderate gut – Band 45-60 x 1.0 = 45-60, midpoint 52.5, cap 70 → **52.5 g/hr x 2h = 105g** *Example (cycle): 3hr ride, high gut* – Band 60-90 x 1.2 = 72-108, midpoint 90, cap 120 → **90 g/hr x 3h = 270g** *Example (swim): 45min swim, any gut* – Ceiling = 0 → **0g during** (fuel in before-phase)

Hydration & Sodium

Sweat rate [Baker 2017]: Light=0.75, Medium=1.25, Heavy=2.0 L/hr. **NO JUSTIFICATION** for exact tier values – Baker provides continuous data, not these tiers.

Temp adjustment: $\text{adjusted} = \text{base} \times (1.0 + \max(0, (\text{temp}_C - 20) \times 0.04))$. **NO JUSTIFICATION** for 0.04 coefficient.

Sodium concentration [Baker 2016]: Low=550, Med=925, High=1150 mg/L. **NO JUSTIFICATION** for exact tier values – Baker provides continuous distribution (200-2000 mg/L), not these tiers.

Formula	Rate	Source
Sodium	$\text{sweat_rate} \times \text{sodium_conc} \times 0.60$	[Blog]: 50-80% range. 60% is our choice.
Hydration	$\text{sweat_rate} \times 1000 \times 0.75$	[Sawka 2007]: ~80%. 75% is our choice.

Sport applicability: Hydration and sodium formulas above apply identically to running and cycling — sweat rate categories, environmental adjustments, and replacement percentages (75% hydration, 60% sodium) are not sport-differentiated. **NO JUSTIFICATION** for using identical hydration parameters across running and cycling — [Baker 2017] studied 506 athletes across sports but does not validate sport-agnostic application of our specific tier values. Swimming receives **0 ml hydration and 0 mg sodium** during the activity (physically impossible to drink while swimming).

3. AFTER Workout

Macro	Formula	Source
Carbs	$\text{wt} \times 1.0 \text{ g/kg} (< 2\text{h}) \text{ or } 1.2 \text{ g/kg} (>= 2\text{h})$	[Blog, Kerksick 2017]: 1.0-1.2 g/kg/hr
Carbs (fasted)	x1.2 boost	NO JUSTIFICATION
Protein	$\text{wt} \times 0.30 \text{ g/kg (fed) or } 0.35 \text{ g/kg (fasted)}$	[Blog, Moore 2009]: 0.25-0.4 g/kg
Fat	$\text{wt} \times 0.20 \text{ g/kg}$	NO JUSTIFICATION

Post-workout hydration:

Macro	Formula	Source
Sodium	$\text{clamp}(\text{deficit} \times 0.50, 300, 700) \text{ mg}$	NO JUSTIFICATION for 50% factor and 300-700 clamp
Hydration	$\text{max}(500, \text{deficit} \times 1.50) \text{ ml}$	[Shirreffs 1996]: 150% of losses. 500ml min: NO JUST.

4. BRICK Workouts (Multi-Sport)

All Before/During/After formulas apply per-segment. Additional rules:

Brick penalty: Run after bike = carb rate x **0.80** (20% reduction) [[Blog](#): 20-30%, [R5: Van Wijck 2012](#)]

Transitions – NO JUSTIFICATION for transition values. T1/T2 carb rates and all duration thresholds are Mealvana’s own.

Total Duration	T1 Carbs	T2 Carbs	Na (T1/T2)	Water (T1/T2)
< 90 min	0	0	0 / 0	0 / 0
90-180 min	0	0	0 / 0	50 / 50
180-420 min	wt x 0.30	wt x 0.35	150 / 100	150 / 100
420+ min	wt x 0.30	wt x 0.35	200 / 150	200 / 150

Swimming: 0 nutrition during (cannot eat/drink). [[Blog](#)]

5. Energy & Environment

Running MET (ACSM): $V02 = 0.2 \times \text{speed_m/min} + 3.5$ (run) or $0.1 \times \text{speed} + 3.5$ (walk). $\text{MET} = V02 / 3.5$ [[ACSM](#)]

Gross kcal = $\text{MET} \times 3.5 \times \text{weight_kg} / 200 \times \text{duration_min}$ [[ACSM](#)]

Net kcal: Running = $1.0 \times \text{wt} \times \text{dist_km}$ [[Margaria 1963](#)] | Cycling = $\text{wt} \times \text{dist} \times 0.3\text{--}0.5$ **NO JUST.** | Swimming = $3.5 \times \text{wt} \times \text{dist_km}$ **NO JUST.**

Environment – NO JUSTIFICATION for exact breakpoints and multiplier values. Informed by [[Sawka 2007](#)] and [[Baker 2017](#)] but tiers are Mealvana’s own.

Condition	Multiplier	Condition	Multiplier
<= 10C	0.85 (cool)	<= 30C or 75-85%	1.2 (hot)
<= 20C, <= 60%	1.0 (temperate)	> 30C or > 85%	1.3 (very hot)
<= 25C or 60-75%	1.1 (warm)		

APPENDIX A: Rachel's Consultation Notes

Rachel Note #1: Pre-Workout Carb Calculation

Recommendation: x g/kg for x hours of pre-workout window (e.g., 3-hour window = 3 g/kg total)

Key clarification: The total amount scales with time, but gets *distributed* across eating opportunities – not stacked/cumulated at each window.

Implementation: Pre-workout carb calculation is directly linked to pre-workout window length (app already does this correctly).

v3 error: Calculated each sub-window separately and added them together, resulting in excessive carb recommendations. Should be one total distributed across windows.

Rachel Note #2: Intensity Affects Window, Not g/kg

Recommendation: Harder/longer workouts -> longer ideal pre-workout window -> more total carbs (because more hours). But g/kg per hour stays constant at ~1.

Example: If you wake up 30 min before a hard workout, the recommendation is still just ~0.5 g/kg top-off – intensity doesn't change that math.

Implementation: Suggest an ideal pre-workout window based on intensity/duration, clearly marked as “(ideal)”. Allow user to select a different window based on their schedule. Calculate carbs based on the window they choose. **v3 error:** Used intensity as a multiplier in the carb calculation. Remove this – intensity only influences the suggested window, not the g/kg formula.

Rachel Note #3: During Carbs Are Absolute, Not Body-Weight Dependent

Recommendation: Use g/hr bands based on workout duration, not body weight. Gut absorption capacity doesn't scale with body mass.

Research-based bands (for moderate gut training): Refer to v3 Table 1

Implementation: Remove any body-weight scaling from during-workout carb calculations. These bands represent the 1.0x baseline for moderately gut-trained athletes.

Rachel Note #4: Gut Training Multipliers

Recommendation: Gut training level scales the entire band up or down – not just where you fall within a fixed band.

Multipliers:

Level	Factor	Effect
Low gut training	0.7x	Reduced capacity, scale band down
Moderate gut training	1.0x	Baseline, use research bands as-is
High gut training	1.2x	Expanded capacity, scale band up

Example for 120-min workout (base band: 45-60 g/hr):

Level	Factor	Result
Low gut	0.7x	31-42 g/hr
Moderate	1.0x	45-60 g/hr
High gut	1.2x	54-72 g/hr

Implementation: Apply multiplier to both ends of the band, preserving the range. Rachel suggested display as a range (e.g., “54-72 g/hr”) not a single number; however, we may need to use the target for now. **v3 error:** Used hard caps instead of multipliers. Remove caps; use scaled ranges.

Rachel Note #5: Blood Diverts from Gut During Exercise

Van Wijck et al. (2012): "The redistribution of blood flow... significantly reduces blood flow to the gut, leading to hypoperfusion and gastrointestinal compromise."

Ter Steege & Kolkman (2012): "With prolonged duration and/or intensity, the splanchnic blood flow may be decreased by 80% or more."

Mechanism: Sympathetic nervous system activation during exercise causes vasoconstriction in splanchnic organs. Blood redirected to working muscles, heart, lungs, skin. Splanchnic blood flow can decrease 43-80% during maximal exercise.

Algorithm Implication: Supports longer pre-workout windows for high-intensity exercise and the brick penalty (run after bike absorption is reduced).

Rachel's Research Literature Verification

1. Pre-Workout Carb Timing: 1-4 g/kg for 1-4 Hours – VERIFIED

ISSN Position Stand confirms: "glycogen levels are best maintained or increased by consuming high carbohydrate (1-4 g/kg/day) meals or snacks for several hours before commencement of training or competition."

Multiple sources support linear relationship: 1 g/kg at 1 hour, 2 g/kg at 2 hours, up to 4 g/kg at 3-4 hours.

Key Source: Kerkick et al. (2017). ISSN Position Stand: Nutrient Timing.

2. During-Exercise Carbs NOT Body-Weight Dependent – STRONGLY VERIFIED

Jeukendrup (2014): "Carbohydrate intake advice is independent of body weight as well as training status." "There appears to be no correlation between BW and exogenous carbohydrate oxidation. The reason for this lack of correlation is probably that the limiting factor is carbohydrate absorption and absorption is largely independent of BW." "These results clearly show that there is no rationale for expressing carbohydrate recommendations for athletes per kilogram of BW."

Key Source: Jeukendrup A. (2014). "A Step Towards Personalized Sports Nutrition." Sports Medicine, 44(S1), 25-33.

3. Cannot Replenish Glycogen During Exercise – VERIFIED

Recent review (Endocrine Reviews, 2025): "CHO ingestion reduces liver glycogenolysis, preserves blood glucose, and paradoxically accelerates muscle glycogen breakdown."

Consumed carbs during exercise: maintain blood glucose, spare liver glycogen, do NOT replenish muscle glycogen stores.

4. Gut Training Improves Tolerance – VERIFIED

Jeukendrup & colleagues (2022): "While the concept of training with high carbohydrate intakes to improve tolerance seems warranted, it remains to be established whether such practice leads to improved absorption."

Human studies show: "The gut is indeed adaptable and this can be used as a practical method to increase exogenous carbohydrate oxidation."

Key Sources: Jeukendrup (2017). "Training the Gut for Athletes." Costa et al. (2019). "Gut-training: impact of two weeks repetitive gut-challenge."

5. Glycogen Storage Requires Water – VERIFIED

Fernandez-Elias et al. (2015): "Each gram of glycogen is stored in human muscle with at least 3 g of water."

6. Avoid Glycogen-Based Predictions – VALIDATED

High individual variability. Cannot measure real-time glycogen during exercise. Rachel's recommendation to use research-based ranges instead of glycogen predictions is well-supported.

APPENDIX B: Xuan's Notes

There are the main points:

1. **Pre-Workout Carb Calculation: 1 g/kg per Hour of Pre-Workout Window** – Recommendation: x g/kg for x hours of pre-workout window (e.g., 3-hour window = 3 g/kg total). Key clarification: The total amount scales with time, but gets distributed across eating opportunities – not stacked/cumulated at each window. v3 error: Calculated each sub-window separately and added them together, resulting in excessive carb recommendations.
 2. **Intensity Affects Pre-Workout Window, Not g/kg Directly** – Harder/longer workouts -> longer ideal pre-workout window -> more total carbs (because more hours). But g/kg per hour stays constant at ~1. v3 error: Used intensity as a multiplier in the carb calculation. Remove this.
 3. **During-Workout Carbs: Absolute Ranges, Not Body-Weight Dependent** – Use g/hr bands based on workout duration, not body weight. Gut absorption capacity doesn't scale with body mass. Refer to v3 Table 1.
 4. **During-Workout Carbs: Gut Training Applies Multipliers to the Band** – Gut training level scales the entire band up or down. Low (0.7x), Moderate (1.0x), High (1.2x). v3 error: Used hard caps instead of multipliers.
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APPENDIX C: Mealvana Blog Post

“How Mealvana Calculates Your Fueling: The Science Behind Your Personalized Nutrition Plan” Published: January 21, 2026 | URL: <https://endurance.mealvana.io/blog/2026/01/21/how-mealvana-calculates-race-fueling/>

Duration-Based Carbohydrate Targets

Duration	Target	Rationale
<60 min	0-30 g total	Mouth rinse sufficient
60-90 min	30-60 g/hr	Maintain blood glucose
90 min - 2.5 hr	45-60 g/hr	Delay glycogen depletion
2.5 - 4 hr	60-90 g/hr	Multiple transportable carbs required
4+ hr	80-100+ g/hr	Maximum sustainable with real food

Pre-Workout Timing Windows

Timing	Carbs (g/kg)	Fat	Fiber	Notes
3-4 hr before	1-4	0.3-0.5 g/kg	Moderate	Full gastric emptying
1-2 hr before	1-2	<10g	Low	Snack window
30-60 min before	0.5-1	0g	0g	Simple carbs only
<30 min before	Not recommended	0g	0g	Emergency only (25-50g)

Sport-Specific Differences

Sport	Practical Carb Ceiling	Pre-Workout Buffer	Rationale
Running	50-70 g/hr	3-4 hours minimum	GI jostling, vertical impact
Cycling	80-120 g/hr	2-3 hours acceptable	Stable platform, no impact
Swimming	Pre/post only	2-3 hours minimum	Horizontal position, no during-activity fueling

Sodium Personalization

Sweat sodium concentration ranges from 200-2,000 mg/L (10x variation). Sodium Loss (mg/hr) = Sweat Rate (L/hr) x Sodium Concentration (mg/L). Target 50-80% replacement during exercise. Hot conditions: 300-600 mg/hr minimum.

Post-Workout Recovery

Goal	Timing	Target	Evidence
Glycogen resynthesis	0-30 min, then hourly	1.0-1.2 g/kg/hr	Maximal 4 hours post
Muscle protein synthesis	Within 2 hours	0.25-0.4 g/kg (20-40g)	~3g leucine threshold
Rehydration	2-4 hours	150% of fluid losses	With 500-700 mg/L sodium

Brick Workouts

“20-30% reduction in run carb targets after bike leg”

Key Quotes

On carbohydrate absorption: "Single carbohydrate sources max out at ~60 g/hr due to SGLT1 transporter saturation." "Glucose + fructose (using GLUT5 transporter) enables 90-120 g/hr absorption."

On intensity: "Duration, intensity (TSS/IF), sport type, environmental conditions, personal tolerance, and training context all influence recommendations simultaneously."

On gut training: "Athletes who consistently practice high carbohydrate intake during training upregulate intestinal transporters (SGLT1 and GLUT5), enabling higher absorption rates."

APPENDIX D: Published Research References

Kerksick et al. (2017) – ISSN Position Stand: Nutrient Timing

International Society of Sports Nutrition Position Stand. Confirms pre-workout 1-4 g/kg for 1-4 hours. Post-workout 1.0-1.2 g/kg/hr for glycogen resynthesis. PubMed: <https://pubmed.ncbi.nlm.nih.gov/28919842/>

Jeukendrup (2014) – Personalized Sports Nutrition

“A Step Towards Personalized Sports Nutrition: Carbohydrate Intake During Exercise.” Sports Medicine, 44(S1), 25-33. Establishes that during-exercise carbohydrate recommendations are independent of body weight. PubMed: <https://pubmed.ncbi.nlm.nih.gov/24791914/>

Jeukendrup (2004) – Carbohydrate Intake During Exercise

Established the 60 g/hr ceiling for single carbohydrate sources via SGLT1 transporter. PubMed: <https://pubmed.ncbi.nlm.nih.gov/15212750/>

Jeukendrup (2017) – Training the Gut for Athletes

“Training the Gut for Athletes.” Sports Medicine, 47(S1), 101-110. Evidence that gut is adaptable and gut training can increase exogenous carbohydrate oxidation. PubMed: <https://pubmed.ncbi.nlm.nih.gov/28508671/>

Costa et al. (2019) – Gut Training

“Gut-training: impact of two weeks repetitive gut-challenge.” 2-week gut training protocols show measurable improvements in tolerance.

Baker et al. (2016-2017) – Sweat Sodium Variability

Studied 506 athletes. Sweat sodium concentration: 230-1,840 mg/L. Established the wide individual variation in sweat composition. PubMed: <https://pubmed.ncbi.nlm.nih.gov/26553489/> (2016), <https://pubmed.ncbi.nlm.nih.gov/28332114/> (2017)

Sawka et al. (2007) – ACSM Position Stand: Exercise and Fluid Replacement

ACSM Position Stand. Recommends 5-7 ml/kg pre-exercise hydration 4 hours before. During exercise: match ~80% of sweat losses. PubMed: <https://pubmed.ncbi.nlm.nih.gov/17277604/>

Moore et al. (2009) – Protein Dose-Response

Established 20g protein plateau for muscle protein synthesis. Supports 0.25-0.4 g/kg range for recovery. PubMed: <https://pubmed.ncbi.nlm.nih.gov/19056590/>

Shirreffs et al. (1996) – Post-Exercise Rehydration

Established that 150% of fluid losses needed for complete rehydration due to ongoing renal losses. PubMed: <https://pubmed.ncbi.nlm.nih.gov/8897382/>

Pfeiffer et al. (2012) – GI Problems in Endurance Sports

Documented gastrointestinal problems vary by sport type. Running has higher GI distress than cycling due to mechanical jostling. PubMed: <https://pubmed.ncbi.nlm.nih.gov/21775906/>

Van Wijck et al. (2012) – Splanchnic Hypoperfusion

“Physiology and pathophysiology of splanchnic hypoperfusion and intestinal injury during exercise.” Splanchnic blood flow reduced up to 80% during intense exercise. PubMed: <https://pubmed.ncbi.nlm.nih.gov/22253445/>

Margaria et al. (1963) – Running Energy Cost

Established running energy cost of approximately 1 kcal/kg/km, independent of speed.

ACSM Metabolic Equations

Standard ACSM metabolic equations for exercise prescription. Running VO₂: $0.2 \times \text{speed (m/min)} + 3.5$. Walking VO₂: $0.1 \times \text{speed (m/min)} + 3.5$. Gross kcal: $\text{MET} \times 3.5 \times \text{weight} / 200 \times \text{duration}$.